UNIT - IV

Laplace Transforms

(Definition, standard transforms, shifting theorems, derivatives & integrals, unit step, convolution, periodic functions)

1. Definition & Existence

Let $f(t)$ be defined for $t \geq 0$. The LAPLACE TRANSFORM of $f(t)$ is

Definition

$$L\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t)\, dt$$

- s is a real (or complex) parameter; the integral must converge.
  $f(t)$ is the INVERSE transform: $f(t) = L^{-1}\{F(s)\}$.
- Laplace transform converts a differential equation in t into an ALGEBRAIC equation in s - very useful for circuits, vibrations, control.

Existence theorem (sufficient conditions)

- $f(t)$ is PIECEWISE CONTINUOUS on every finite interval $[0, b]$ (finite jumps only).
- $f(t)$ is of EXPONENTIAL ORDER a: $|f(t)| \leq Me^{at}$ for all $t > T$. Then $L\{f\}$ exists for $s > a$.

These are sufficient, not necessary: e.g. $f = t^{-1/2}$ is not bounded at 0 but its transform exists. Functions like e^{t^2} or $\tan t$ have NO Laplace transform.

Linearity

$$L\{af(t)+bg(t)\}=aL\{f(t)\}+bL\{g(t)\}$$

Derivation of basic transforms

(i) $L\{1\}=\int_0^\infty e^{-st}dt=\left[\frac{e^{-st}}{-s}\right]_0^\infty=\frac{1}{s},\ s>0.$

(ii) $L\{e^{at}\}=\int_0^\infty e^{-(s-a)t}dt=\frac{1}{s-a},\ s>a.$

(iii) $L\{t^n\}=\int_0^\infty e^{-st}t^n dt=\frac{\Gamma(n+1)}{s^{n+1}}=\frac{n!}{s^{n+1}}$ (put $st=x$).

(iv) $L\{\sin at\}$: $L\{e^{iat}\}=\frac{1}{s-ia}=\frac{s+ia}{s^2+a^2}$; equate real and imaginary parts $\Rightarrow L\{\cos at\}=\frac{s}{s^2+a^2}, L\{\sin at\}=\frac{a}{s^2+a^2}.$

2. Laplace Transforms of Standard Functions

$f(t)$	$F(s)=L\{f\}$	$f(t)$	$F(s)$
1	$1/s$	$\sinh at$	$a/(s^2-a^2)$
t^n (n integer)	$n!/s^{n+1}$	$\cosh at$	$s/(s^2-a^2)$
e^{at}	$1/(s-a)$	te^{at}	$1/(s-a)^2$
$\sin at$	$a/(s^2+a^2)$	$e^{at}\sin bt$	$b/[(s-a)^2+b^2]$
$\cos at$	$s/(s^2+a^2)$	$e^{at}\cos bt$	$(s-a)/[(s-a)^2+b^2]$
$t\sin at$	$2as/(s^2+a^2)^2$	$t\cos at$	$(s^2-a^2)/(s^2+a^2)^2$
$t^{-1/2}$	$\sqrt{\pi/s}$	$t^{1/2}$	$\sqrt{\pi}/(2s^{3/2})$

Example 1. Find the Laplace transforms

1. $L\{3t^2-4t+5+2e^{-3t}\}=\frac{6}{s^3}-\frac{4}{s^2}+\frac{5}{s}+\frac{2}{s+3}$

2. $\sin 2t\cos 3t=\frac{1}{2}(\sin 5t-\sin t)\ \Rightarrow L=\frac{1}{2}\left[\frac{5}{s^2+25}-\frac{1}{s^2+1}\right]$

3. $\cos^2 t=\frac{1}{2}(1+\cos 2t)\ \Rightarrow L=\frac{1}{2}\left[\frac{1}{s}+\frac{s}{s^2+4}\right]=\frac{s^2+2}{s(s^2+4)}$

4. $\sin^3 t=\frac{1}{4}(3\sin t-\sin 3t)\ \Rightarrow$

$$L=\frac{1}{4}\left[\frac{3}{s^2+1}-\frac{3}{s^2+9}\right]=\frac{6}{(s^2+1)(s^2+9)}$$

5. $\cosh^2 2t = \frac{1}{2}(1+\cosh 4t) \Rightarrow L = \frac{1}{2}\left[\frac{1}{s} + \frac{s}{s^2-16}\right]$

3. Properties of Laplace Transforms

(a) First Shifting Theorem (shift in s)

First shifting theorem

If $L\{f(t)\} = F(s)$ *then*

$$L\{e^{at} f(t)\} = F(s-a)$$

i.e. replace s *by* $s-a$ *in* $F(s)$ *(and by* $s+a$ *for* e^{-at}*).*

Proof: $L\{e^{at} f\} = \int_0^\infty e^{-(s-a)t} f(t)dt = F(s-a).$

Example 2.

1. $L\{e^{2t}\cos 3t\} = \dfrac{s-2}{(s-2)^2+9}$
2. $L\{e^{-t}t^3\} = \dfrac{3!}{(s+1)^4} = \dfrac{6}{(s+1)^4}$
3. $L\{e^{-3t}\sin 2t\} = \dfrac{2}{(s+3)^2+4}$
4. $L\{e^t(3\cosh 2t - 5\sinh 2t)\} = \dfrac{3(s-1)-10}{(s-1)^2-4} = \dfrac{3s-13}{s^2-2s-3}$

(b) Change of scale

$$L\{f(at)\} = \frac{1}{a} F\left(\frac{s}{a}\right), \quad a > 0$$

(c) Multiplication by t^n (derivative of the transform)

$$L\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} F(s)$$

Example 3.

1. $L\{t\sin 2t\} = -\dfrac{d}{ds}\left(\dfrac{2}{s^2+4}\right) = \dfrac{4s}{(s^2+4)^2}$
2. $L\{t^2\cos t\} = \dfrac{d^2}{ds^2}\left(\dfrac{s}{s^2+1}\right) = \dfrac{2s^3-6s}{(s^2+1)^3}$
3. $L\{te^{-t}\sin t\}$: *first* $L\{t\sin t\} = \dfrac{2s}{(s^2+1)^2}$*; shift* $s \to s+1$*:*

$$\frac{2(s+1)}{[(s+1)^2+1]^2}$$

(d) Division by t

$$L\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(u)\,du$$

1. $L\left\{\dfrac{\sin t}{t}\right\} = \displaystyle\int_s^\infty \frac{du}{u^2+1} = \tan^{-1}\frac{1}{s} = \cot^{-1} s$
2. $L\left\{\dfrac{1-\cos t}{t}\right\} = \displaystyle\int_s^\infty \left(\frac{1}{u} - \frac{u}{u^2+1}\right)du = \frac{1}{2}\ln\frac{s^2+1}{s^2}$
3. $L\left\{\dfrac{e^{-at}-e^{-bt}}{t}\right\} = \ln\dfrac{s+b}{s+a}$

Properties at a glance

Operation on $f(t)$	Effect on $F(s)$
multiply by e^{at}	*shift:* $F(s-a)$
multiply by t^n	$(-1)^n F^{(n)}(s)$
divide by t	$\displaystyle\int_s^\infty F(u)du$
differentiate	$sF(s) - f(0)$
integrate $0 \to t$	$F(s)/s$
delay by a: $f(t-a)u(t-a)$	$e^{-as}F(s)$
convolution $f * g$	$F(s)\,G(s)$

4. Transforms of Derivatives and Integrals

Derivatives

$$L\{f'(t)\} = sF(s) - f(0)$$

$$L\{f''(t)\} = s^2F(s) - s\,f(0) - f'(0)$$

$$L\{f^{(n)}(t)\} = s^nF(s) - s^{n-1}f(0) - s^{n-2}f'(0) - \cdots - f^{(n-1)}(0)$$

Proof for f' (integration by parts):

$$\int_0^\infty e^{-st}f'dt = [e^{-st}f]_0^\infty + s\int_0^\infty e^{-st}f\,dt = -f(0) + sF(s).$$

Integrals

$$L\left(\int_0^t f(u)\,du\right)=\frac{F(s)}{s}$$

Example 4.

1. *Find* $L\{\sin^2 t\}$ *using derivatives. Let* $f=\sin^2 t$, $f(0)=0$, $f'=\sin 2t$. *Then* $L\{f'\}=sF$ *gives* $\dfrac{2}{s^2+4}=sF\Rightarrow F=\dfrac{2}{s(s^2+4)}$.
2. $L\left(\displaystyle\int_0^t \frac{\sin u}{u}\,du\right)=\frac{1}{s}\tan^{-1}\frac{1}{s}$
3. $L\left(\displaystyle\int_0^t e^{-u}\cos u\,du\right)=\frac{1}{s}\cdot\frac{s+1}{(s+1)^2+1}$

5. Inverse Laplace Transform

If $L\{f(t)\}=F(s)$ *then* $f(t)=L^{-1}\{F(s)\}$. *Inverse is also linear. Basic list (read the table of Section 2 backwards):*

$F(s)$	$L^{-1}\{F\}$	$F(s)$	$L^{-1}\{F\}$
$1/s$	1	$1/(s^2+a^2)$	$\frac{1}{a}\sin at$
$1/s^n$	$t^{n-1}/(n-1)!$	$s/(s^2+a^2)$	$\cos at$
$1/(s-a)$	e^{at}	$1/(s^2-a^2)$	$\frac{1}{a}\sinh at$
$s/(s^2-a^2)$	$\cosh at$	$s/(s^2+a^2)^2$	$\frac{t\sin at}{2a}$

Methods: (1) partial fractions, (2) completing the square + first shift, (3) first shifting theorem, (4) convolution theorem, (5) derivative/integral rules.

Example 5. Find the inverse Laplace transforms

1. $\dfrac{3s+7}{s^2-2s-3}=\dfrac{3s+7}{(s-3)(s+1)}=\dfrac{4}{s-3}-\dfrac{1}{s+1}\Rightarrow f(t)=4e^{3t}-e^{-t}$
2. $\dfrac{s+2}{s^2+4s+13}=\dfrac{s+2}{(s+2)^2+3^2}\Rightarrow f(t)=e^{-2t}\cos 3t$
3. $\dfrac{s}{(s^2+1)(s^2+4)}=\dfrac{1}{3}\left[\dfrac{s}{s^2+1}-\dfrac{s}{s^2+4}\right]\Rightarrow f(t)=\dfrac{1}{3}(\cos t-\cos 2t)$

4. $\frac{1}{s^2(s^2+1)} = \frac{1}{s^2} - \frac{1}{s^2+1} \Rightarrow f(t) = t - \sin t$
5. $\frac{2s+3}{s^2+6s+25} = \frac{2(s+3)-3}{(s+3)^2+16} \Rightarrow f(t) = e^{-3t}\left(2\cos 4t - \frac{3}{4}\sin 4t\right)$
6. $\ln\frac{s+1}{s-1}$: $F'(s) = \frac{1}{s+1} - \frac{1}{s-1} \Rightarrow L^{-1}\{F'\} = e^{-t} - e^{t} = -tf(t) \Rightarrow$
   $f(t) = \frac{2\sinh t}{t}$
7. $\frac{1}{s(s+1)^2} = \frac{1}{s} - \frac{1}{s+1} - \frac{1}{(s+1)^2} \Rightarrow f(t) = 1 - e^{-t} - te^{-t}$
8. $\frac{5s+3}{(s-1)(s^2+2s+5)} = \frac{1}{s-1} - \frac{(s+1)-3}{(s+1)^2+4} \Rightarrow$
   $f(t) = e^{t} - e^{-t}\cos 2t + \frac{3}{2}e^{-t}\sin 2t$
9. $\frac{s^2}{(s^2+1)(s^2+4)} = \frac{1}{3}\left[\frac{4}{s^2+4} - \frac{1}{s^2+1}\right] \Rightarrow f(t) = \frac{1}{3}(2\sin 2t - \sin t)$

Evaluation of integrals using Laplace transforms

Put the required value of s in $\int_0^\infty e^{-st} f(t)dt = F(s)$.

1. $\int_0^\infty te^{-3t}\sin t\, dt = L\{t\sin t\}_{s=3} = \frac{2s}{(s^2+1)^2}|_{s=3} = \frac{6}{100} = 0.06$
2. $\int_0^\infty \frac{e^{-t} - e^{-2t}}{t} dt = \ln\frac{s+2}{s+1}|_{s=0} = \ln 2$
3. $\int_0^\infty \frac{\sin t}{t} dt = \cot^{-1} s|_{s=0} = \frac{\pi}{2}$
4. $\int_0^\infty e^{-2t}\cos 3t\, dt = \frac{s}{s^2+9}|_{s=2} = \frac{2}{13}$

Initial and final value theorems (useful facts)

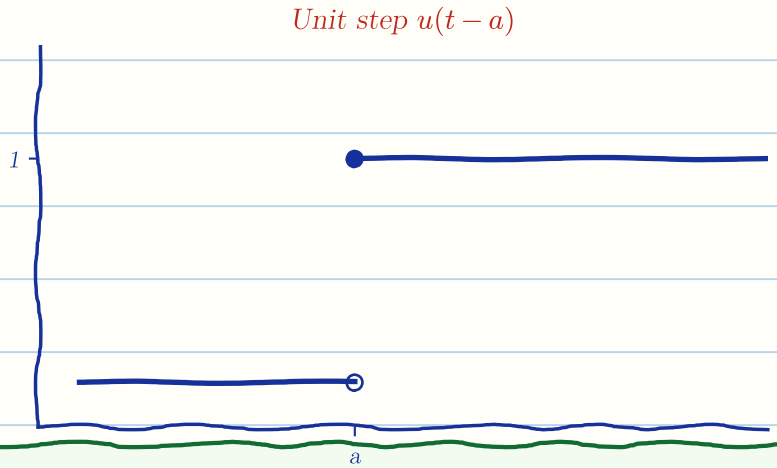
$$f(0^+) = \lim_{s\to\infty} sF(s), \qquad f(\infty) = \lim_{s\to 0} sF(s)$$

Example: $F(s) = \frac{3}{s(s+2)}$: $f(0) = \lim 3/(s+2) = 0$; $f(\infty) = 3/2$.

(Check: $f = \frac{3}{2}(1 - e^{-2t})$.*)*

6. Unit Step (Heaviside) Function & Second Shifting Theorem

The UNIT STEP FUNCTION at $t=a$: $u(t-a)=0$ for $t<a$ and $u(t-a)=1$ for $t\geq a$.



Second shifting theorem

$$L\{u(t-a)\}=\frac{e^{-as}}{s}$$

$$L\{f(t-a)\,u(t-a)\}=e^{-as}F(s) \quad \text{(Second shifting theorem)}$$

$$L^{-1}\{e^{-as}F(s)\}=f(t-a)\,u(t-a)$$

Alternative form: $L\{g(t)\,u(t-a)\}=e^{-as}L\{g(t+a)\}$ (shift the function first).

Example 6.

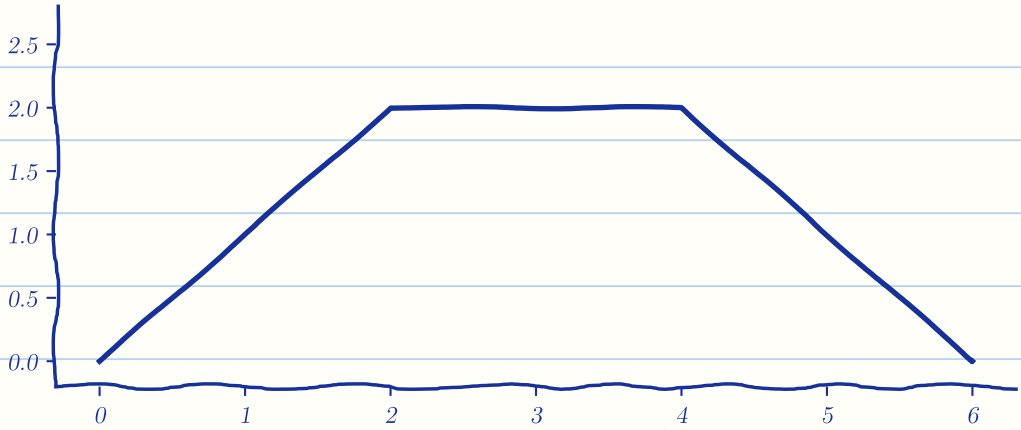
1. $L\{(t-2)^3u(t-2)\}=e^{-2s}\frac{3!}{s^4}=\frac{6e^{-2s}}{s^4}$
2. $f(t)=t^2$ *for* $t>2$, 0 *otherwise.* $f=t^2u(t-2)$;
   $g(t+2)=(t+2)^2=t^2+4t+4 \Rightarrow L\{f\}=e^{-2s}\left[\frac{2}{s^3}+\frac{4}{s^2}+\frac{4}{s}\right]$
3. $f(t)=\sin t$ *for* $0<t<\pi$, 0 *for* $t>\pi$.
   $f=\sin t-\sin t\,u(t-\pi)=\sin t+\sin(t-\pi)u(t-\pi) \Rightarrow$
   $L\{f\}=\frac{1+e^{-\pi s}}{s^2+1}$

4. $L^{-1}\left\{\dfrac{e^{-3s}}{s^2+4}\right\}=\dfrac{1}{2}\sin 2(t-3)\,u(t-3)$

5. $L^{-1}\left\{\dfrac{e^{-\pi s}}{s(s^2+1)}\right\}: L^{-1}\left\{\frac{1}{s(s^2+1)}\right\}=1-\cos t \;\Rightarrow$

$[1-\cos(t-\pi)]\,u(t-\pi)=(1+\cos t)\,u(t-\pi)$

$f(t)=t[u(t)-u(t-2)]+2[u(t-2)-u(t-4)]+(6-t)u(t-4)$



A piecewise function is written with step functions: each new piece is switched ON by $u(t-a)$, and the old one switched OFF.

7. Convolution Theorem

The CONVOLUTION of f and g is $(f*g)(t)=\displaystyle\int_0^t f(u)\,g(t-u)\,du$,

with $f*g=g*f$.

Convolution theorem

$$L\{f*g\}=L\{f\}\cdot L\{g\}=F(s)\,G(s)$$

$$L^{-1}\{F(s)\,G(s)\}=\int_0^t f(u)\,g(t-u)\,du$$

Example 7. Use convolution to find

1. $L^{-1}\left\{\dfrac{1}{s(s^2+1)}\right\}=1*\sin t=\displaystyle\int_0^t \sin u\,du=1-\cos t$

2. $L^{-1}\left\{\dfrac{1}{(s+1)(s-2)}\right\}=e^{-t}*e^{2t}=\displaystyle\int_0^t e^{-u}e^{2(t-u)}du=\frac{e^{2t}-e^{-t}}{3}$

3. $L^{-1}\left\{\dfrac{1}{s^2(s^2+1)}\right\}=t*\sin t=\displaystyle\int_0^t u\sin(t-u)\,du=t-\sin t$

4. $L^{-1}\left\{\dfrac{s}{(s^2+a^2)^2}\right\} = \cos at * \dfrac{\sin at}{a} = \dfrac{t\sin at}{2a}$

5. $L^{-1}\left\{\dfrac{1}{(s^2+a^2)^2}\right\} = \dfrac{\sin at}{a} * \dfrac{\sin at}{a} = \dfrac{\sin at - at\cos at}{2a^3}$

6. $L^{-1}\left\{\dfrac{s^2}{(s^2+4)^2}\right\} = \cos 2t * \cos 2t \ \ldots \ = \dfrac{1}{4}\sin 2t + \dfrac{t}{2}\cos 2t$

8. Laplace Transform of Periodic Functions

If $f(t+T) = f(t)$ for all t (period T), then

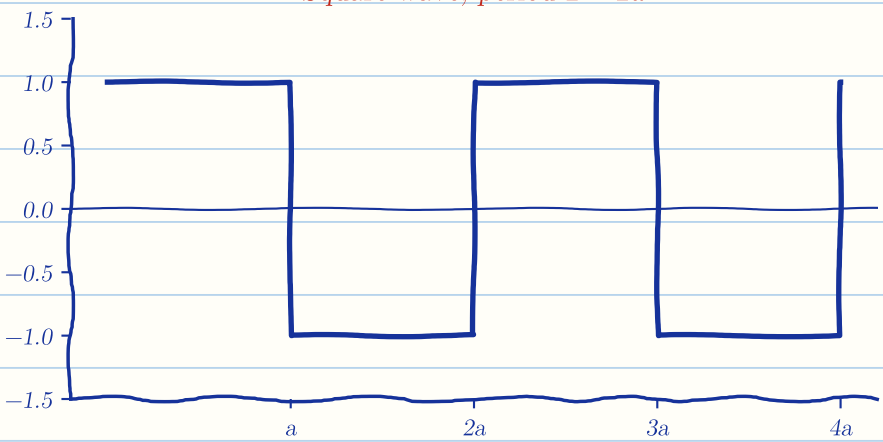
Periodic function formula

$$L\{f(t)\} = \frac{1}{1-e^{-sT}} \int_0^T e^{-st} f(t)\, dt$$

Proof: split $\int_0^\infty$ *into periods, put* $t = u + nT$, *sum the geometric series*

$$\sum e^{-nsT} = \frac{1}{1-e^{-sT}}.$$

Square wave, period $T = 2a$



Example 8.

1. *SQUARE WAVE:* $f = 1$ $(0 < t < a)$, $f = -1$ $(a < t < 2a)$, $T = 2a$.

$$\int_0^{2a} e^{-st} f\, dt = \frac{(1-e^{-as})^2}{s}$$

$$L\{f\} = \frac{(1-e^{-as})^2}{s(1-e^{-2as})} = \frac{1-e^{-as}}{s(1+e^{-as})} = \frac{1}{s}\tanh\frac{as}{2}$$

2. *SAWTOOTH:* $f = t$ $(0 < t < a)$, *period* a.

$$\int_0^a te^{-st}\, dt = \frac{1}{s^2} - \frac{ae^{-as}}{s} - \frac{e^{-as}}{s^2}$$

$$L\{f\} = \frac{1}{s^2} - \frac{a\, e^{-as}}{s(1-e^{-as})}$$

3. *FULL-WAVE RECTIFIED SINE* $f=|\sin t|$, *period* π:

$$\int_0^{\pi} e^{-st}\sin t\,dt=\frac{1+e^{-\pi s}}{s^2+1}$$

$$L\{|\sin t|\}=\frac{1+e^{-\pi s}}{(s^2+1)(1-e^{-\pi s})}=\frac{1}{s^2+1}\coth\frac{\pi s}{2}$$

4. *HALF-WAVE RECTIFIED sine* (\$\sin t\$ *for* $0<t<\pi$, 0 *for* $\pi<t<2\pi$, $T=2\pi$): $L\{f\}=\dfrac{1}{(s^2+1)(1-e^{-\pi s})}$

5. *TRIANGULAR WAVE:* $f=t$ $(0<t<a)$, $f=2a-t$ $(a<t<2a)$, $T=2a$: $L\{f\}=\dfrac{1}{s^2}\tanh\dfrac{as}{2}$

(*Since* $\displaystyle\int_0^{2a} e^{-st}f\,dt=\frac{(1-e^{-as})^2}{s^2}$ *and* $1-e^{-2as}=(1-e^{-as})(1+e^{-as})$.)

9. Solving Differential Equations (application)

Procedure: (1) take L *of both sides using derivative formulae and initial conditions; (2) solve algebraically for* $Y(s)$; *(3) take* L^{-1} *(partial fractions).*

Example 9. $y''+4y'+3y=e^{-t}$, $y(0)=0$, $y'(0)=0$

$$[s^2Y]+4[sY]+3Y=\frac{1}{s+1}\ \Rightarrow$$

$$Y=\frac{1}{(s+1)^2(s+3)}=\frac{-1/4}{s+1}+\frac{1/2}{(s+1)^2}+\frac{1/4}{s+3}$$

$$y(t)=-\frac{1}{4}e^{-t}+\frac{1}{2}te^{-t}+\frac{1}{4}e^{-3t}$$

Check: $y(0)=-\frac{1}{4}+\frac{1}{4}=0$, $y'(0)=\frac{1}{4}+\frac{1}{2}-\frac{3}{4}=0$. *OK.*

Example 10. $y''+y=\sin 2t$, $y(0)=y'(0)=0$

$$(s^2+1)Y=\frac{2}{s^2+4}\ \Rightarrow Y=\frac{2}{(s^2+1)(s^2+4)}=\frac{2}{3}\left[\frac{1}{s^2+1}-\frac{1}{s^2+4}\right]\Rightarrow$$

$$y=\frac{2}{3}\sin t-\frac{1}{3}\sin 2t$$

Example 11. $y' + 2y = u(t-1),\ y(0) = 0$

$(s+2)Y = \frac{e^{-s}}{s} \Rightarrow Y = e^{-s} \cdot \frac{1}{2}\left[\frac{1}{s} - \frac{1}{s+2}\right] \Rightarrow$

$y(t) = \frac{1}{2}(1 - e^{-2(t-1)})u(t-1)$

Example 12. $y'' + 9y = \cos 2t,\ y(0) = 1,\ y'(0) = 0$

$(s^2Y - s) + 9Y = \frac{s}{s^2+4} \Rightarrow$

$Y = \frac{s}{s^2+9} + \frac{s}{(s^2+4)(s^2+9)} = \frac{s}{s^2+9} + \frac{1}{5}\left[\frac{s}{s^2+4} - \frac{s}{s^2+9}\right]$

$$y(t) = \frac{4}{5}\cos 3t + \frac{1}{5}\cos 2t$$

Example 13. $y'' + 4y = u(t-2),\ y(0) = y'(0) = 0$ **(switch-on force)**

$Y = \frac{e^{-2s}}{s(s^2+4)}$, and $\frac{1}{s(s^2+4)} = \frac{1}{4}\left[\frac{1}{s} - \frac{s}{s^2+4}\right] \Rightarrow$

$y(t) = \frac{1}{4}[1 - \cos 2(t-2)]u(t-2)$

Impulse (Dirac delta) - extra

$\delta(t-a)$ *is a unit impulse at* $t = a$ *with* $\int f(t)\delta(t-a)dt = f(a)$ *and* $L\{\delta(t-a)\} = e^{-as}$. *Example:* $y'' + y = \delta(t-\pi),\ y(0) = y'(0) = 0 \Rightarrow$

$Y = \frac{e^{-\pi s}}{s^2+1} \Rightarrow y = \sin(t-\pi)\,u(t-\pi) = -\sin t\,u(t-\pi).$

10. Quick Revision Sheet

Formulae

$L\{t^n\} = \frac{n!}{s^{n+1}},\ L\{e^{at}\} = \frac{1}{s-a},\ L\{\sin at\} = \frac{a}{s^2+a^2},$
$L\{\cos at\} = \frac{s}{s^2+a^2}$
$L\{e^{at}f\} = F(s-a);\ L\{t^n f\} = (-1)^n F^{(n)}(s)\ ;\ L\{f/t\} = \int_s^\infty F\,du$
$L\{f'\} = sF - f(0)\ ;\ L\{f''\} = s^2F - sf(0) - f'(0)\ ;\ L\{\int f\} = F/s$
$L\{f(t-a)u(t-a)\} = e^{-as}F(s)\ ;\ L\{f*g\} = FG\ ;$ *periodic:* $\frac{\int_0^T e^{-st} f\,dt}{1-e^{-sT}}$

Practice problems (with answers)

1. *Find* $L\{e^{-2t}\sin 3t\cos t\}$. *Ans:* $\frac{2}{(s+2)^2+16} + \frac{1}{(s+2)^2+4}$
2. *Find* $L\{t^2e^{-t}\sin t\}$ *(hint:* $\frac{d^2}{ds^2}$ *of* $L\{e^{-t}\sin t\}$*). Ans:* $\frac{6(s+1)^2-2}{[(s+1)^2+1]^3}$
3. *Find* $L^{-1}\left\{\frac{s+1}{s^2+s+1}\right\}$. *Ans:* $e^{-t/2}\left[\cos\frac{\sqrt{3}}{2}t + \frac{1}{\sqrt{3}}\sin\frac{\sqrt{3}}{2}t\right]$
4. *Solve* $y'' - 3y' + 2y = 4e^{2t},\ y(0) = -3,\ y'(0) = 5$. *Ans:* $y = 4(t+1)e^{2t} - 7e^t$
5. *Find* $L^{-1}\left\{\frac{1}{(s-1)(s^2+4)}\right\}$ *by convolution. Ans:* $\frac{1}{5}\left(e^t - \cos 2t - \frac{1}{2}\sin 2t\right)$

Common mistakes to avoid

- *Forgetting initial conditions in* $L\{y'\}$, $L\{y''\}$. *Using* $s+a$ *instead of* $s-a$ *in the first shift for* e^{at}.
- *Writing* $L\{fg\} = L\{f\}L\{g\}$: *WRONG - product in* t *is NOT product in* s *(only convolution is).*
- *In the second shift the function must be written in terms of* $(t-a)$ *before taking the transform.*
- *For inverse problems: (i) factor denominator, (ii) partial fractions, (iii) complete the square, (iv) shift theorems.*